Exercise Set 17: Index Theory and the Atiyah-Singer Theorem

The Atiyah-Singer index theorem (1963) is the deepest theorem connecting analysis, geometry, and topology. It says: the ***analytical index*** *of a differential operator D (the dimension of its kernel minus the dimension of its cokernel) equals the* ***topological index*** *(an integral of characteristic classes over the manifold). This single theorem unifies Gauss-Bonnet (Sets 6B, 12.5), the Hirzebruch signature theorem (Set 14), the Riemann-Roch theorem (complex geometry), and the Witten index (Set 10) as special cases. In physics it explains why there are zero modes of the Dirac operator, why anomalies exist, and why the path integral sometimes vanishes.*

Recall of Definitions and Examples

R1. Write the definition of a **differential operator** $D : \Gamma(E) \to \Gamma(F)$ between sections of vector bundles $E, F \to M$. What is the **symbol** $\sigma(D)(\xi) : E_x \to F_x$ of D at a point $x \in M$ in direction $\xi \in T_x^*M$? When is D **elliptic**?

R2. Define the **analytical index** of an elliptic operator D:

$$\operatorname{ind}(D) = \dim\ker D - \dim\ker D^* = \dim\ker D - \dim\operatorname{coker} D.$$

Why is this finite for an elliptic operator on a compact manifold?

R3. State the **Atiyah-Singer index theorem**: for an elliptic differential operator D on a compact oriented Riemannian manifold M,

$$\operatorname{ind}(D) = \int_M \hat{A}(TM) \wedge \operatorname{ch}(E/F),$$

where $\hat{A}(TM)$ is the **$\hat{A}$-genus** (a characteristic class built from the Pontryagin classes of TM) and $\operatorname{ch}(E/F)$ is the **Chern character** of the symbol bundle.

R4. State the special cases:

(a) **Gauss-Bonnet-Chern**: $D = d + d^*$, $\operatorname{ind} = \chi(M)$

(b) **Hirzebruch signature**: $D = d + d^*$ (with grading by ± 1 eigenspaces of the Hodge star), $\operatorname{ind} = \sigma(M)$

(c) **Dolbeault (Riemann-Roch)**: $D = \bar{\partial} + \bar{\partial}^*$, $\operatorname{ind} = \chi(\mathcal{O}_M) = \sum(-1)^q h^{0,q}$

(d) **Dirac operator**: $D =$ the Dirac operator, $\operatorname{ind} = \hat{A}(M)$ (the $\hat{A}$-genus)

R5. The **$\hat{A}$-genus** is: $\hat{A}(M) = \int_M \hat{A}(TM)$ where $\hat{A}(TM) = \prod_j \frac{x_j/2}{\sinh(x_j/2)}$ and x_j are the formal Chern roots of TM. Write $\hat{A}$ in terms of Pontryagin classes to first order: $\hat{A}(M) = 1 - \frac{p_1}{24} + \cdots$

R6. The **Chern character** of a bundle E: $\mathrm{ch}(E) = \mathrm{Tr}(e^{iF/2\pi}) = r + c_1 + \frac{c_1^2 - 2c_2}{2} + \cdots$ where $r = \mathrm{rank}(E)$.

Theory

T1. (The de Rham operator and Gauss-Bonnet.)

Let $D = d + d^* : \Omega^{\mathrm{even}}(M) \to \Omega^{\mathrm{odd}}(M)$ where $d^* = (-1)^{nk+n+1} * d *$ on Ω^k.

(a) Show $D^* = d + d^* : \Omega^{\mathrm{odd}} \to \Omega^{\mathrm{even}}$ (same formula, reversed grading).

(b) Show $\ker D = \bigoplus_k \mathcal{H}^{2k}(M)$ (even harmonic forms) by the Hodge theorem.

(c) Compute: $\mathrm{ind}(D) = \sum_k (-1)^k \dim \mathcal{H}^k(M) = \sum_k (-1)^k \dim H^k_{dR}(M) = \chi(M)$.

(d) The Atiyah-Singer theorem in this case: $\chi(M) = \int_M e(TM)$ (the Gauss-Bonnet-Chern theorem). For $\dim M = 2$: $e(TM) = \frac{1}{2\pi} K\, dA$, recovering classical Gauss-Bonnet.

T2. (The signature operator.)

On an oriented Riemannian $4n$-manifold, the Hodge star $* : \Omega^{2n} \to \Omega^{2n}$ has eigenvalues ± 1. Let $\Omega^{2n}_\pm$ be the ± 1 eigenspaces ($\Omega^2_\pm$ from Nawata HW11). The **signature operator** is $D = d + d^* : \Omega^+ \to \Omega^-$ where $\Omega^\pm = \bigoplus_k \Omega^k_\pm$.

(a) Show $\mathrm{ind}(D) = \dim H^{2n}_+ - \dim H^{2n}_- = \sigma(M)$ (the signature of the intersection form).

(b) The Atiyah-Singer theorem: $\sigma(M) = \int_M L(TM)$ where $L(TM) = \prod_j \frac{x_j}{\tanh x_j}$ is the Hirzebruch L-polynomial. For a 4-manifold: $\sigma(M) = \frac{p_1(M)}{3}$ the Hirzebruch signature theorem.

(c) For $\mathbb{CP}^2$: $p_1 = 3$, $\sigma = 1$. Verify.

T3. (The Dirac operator.)

On a spin manifold M (a Riemannian manifold admitting a spin structure), the **Dirac operator** is: $D = \sum_i e_i \cdot \nabla_{e_i} : \Gamma(S^+) \to \Gamma(S^-)$ where $S = S^+ \oplus S^-$ is the spinor bundle and $e_i \cdot$ is Clifford multiplication.

(a) Show $D^2 = \nabla^* \nabla + \frac{R}{4}$ (the **Lichnerowicz formula**), where R is scalar curvature.

(b) Conclude: if $R > 0$ everywhere on a compact spin manifold, then D has no zero modes ($\ker D = 0$).

(c) By Atiyah-Singer: $\mathrm{ind}(D) = \hat{A}(M)$. If $R > 0$, then $\hat{A}(M) = 0$ a purely topological consequence of a geometric condition. This is the **Lichnerowicz obstruction**: a manifold with $\hat{A}(M) \neq 0$ admits no metric of positive scalar curvature.

Computation

C1. (The index of the de Rham operator on specific manifolds.)

(a) $M = S^2$: compute $\chi(S^2) = \int_{S^2} e(TS^2) = \frac{1}{2\pi}\int_{S^2} K\, dA = \frac{1}{2\pi}\cdot 4\pi = 2$. ✓

(b) $M = T^2$: $\chi(T^2) = 0$. Verify from H^k: $\dim H^0 - \dim H^1 + \dim H^2 = 1-2+1 = 0$. ✓

(c) $M = \mathbb{CP}^2$: $\chi(\mathbb{CP}^2) = 3$. Compute from H^k: $1+0+1+0+1 = 3$. ✓

(d) $M = K3$ (from Set 15): $\chi(K3) = 24$. Verify: $h^{p,q}$ with $h^{0,0} = h^{2,2} = h^{2,0} = h^{0,2} = 1$, $h^{1,1} = 20$: $\chi = 2\cdot 1 + 2\cdot 1 + 20 - 0 - 0 = 24$. ✓

C2. (The Witten index as an index theorem.)

From Set 10: the Witten index is $Z = \mathrm{Tr}[(-1)^F e^{-\beta H}] = \chi(M)$.

(a) The Witten index is the analytical index of the operator $D = d + d^* : \Omega^{\text{even}} \to \Omega^{\text{odd}}$. Verify: $\mathrm{ind}(D) = \chi(M) = Z$.

(b) The β-independence of the Witten index is a consequence of the index theorem: the analytical index is a topological invariant, hence independent of any continuous deformation (including β). This explains the result from Set 10 rigorously.

(c) The Morse theory computation of Set 6B gives $Z = \sum_k (-1)^k M_k$ (number of index-k critical points). The Morse inequalities: $M_k \geq b_k$ with $\sum(-1)^k M_k = \sum(-1)^k b_k$. The index theorem says $\sum(-1)^k b_k = \chi(M)$. Show all three computations agree.

C3. ($\hat{A}$-genus and obstructions to positive curvature.)

(a) For a spin 4-manifold: $\hat{A}(M) = -\frac{p_1}{24}$. For $K3$: $p_1(K3) = -48$ (from $\chi = 24$ and $\sigma = -16$ via Noether's formula $\chi + \sigma = 4\cdot(1+p_g)$). Compute $\hat{A}(K3) = -\frac{-48}{24} = 2 \neq 0$. Conclude: $K3$ admits no metric of positive scalar curvature.

(b) For $\mathbb{CP}^2$: $\mathbb{CP}^2$ is not spin ($c_1(\mathbb{CP}^2) = 3x$ is odd spin requires $c_1 \equiv 0 \pmod 2$). So the Dirac operator is not defined. But $\mathbb{CP}^2$ does admit a metric of positive curvature (the Fubini-Study metric has $R > 0$) consistent.

(c) **The power of the theorem**: the Lichnerowicz obstruction $\hat{A}(M) \neq 0$ is computable from characteristic classes alone, with no knowledge of any specific metric. Topology obstructs the existence of positively curved metrics.

C4. (The Dolbeault operator and Riemann-Roch.)

On a compact complex manifold M of complex dimension n, the Dolbeault operator $D = \bar{\partial} + \bar{\partial}^*$ has:

$$\mathrm{ind}(D) = \sum_{q=0}^{n} (-1)^q h^{0,q}(M) = \chi(\mathcal{O}_M)$$

where $\mathcal{O}_M$ is the structure sheaf.

(a) For a compact Riemann surface Σ_g: $h^{0,0} = 1$, $h^{0,1} = g$. Compute $\mathrm{ind}(D) = 1 - g$.

(b) The Riemann-Roch theorem for a line bundle L on Σ_g: $\chi(L) = h^{0,0}(L) - h^{0,1}(L) = \deg(L) + 1 - g$ where $\deg(L) = \int_{\Sigma_g} c_1(L) \in \mathbb{Z}$.

(c) For the Hopf bundle L over $\mathbb{CP}^1 \cong S^2$: $\deg(L) = c_1 = 1$. Riemann-Roch: $\chi(L) = 1 + 1 - 0 = 2$. Verify: $h^{0,0}(L) = 2$ (holomorphic sections of $\mathcal{O}(1)$ are linear polynomials in z: $az_0 + bz_1$), $h^{0,1}(L) = 0$.

Physics: Zero Modes, Anomalies, and Index Theory

The Atiyah-Singer index theorem has three major applications in physics: (1) the number of zero modes of the Dirac operator determines the fermionic content of the path integral; (2) anomalies are index-theoretic in origin the chiral anomaly is the index of the Dirac operator in an external gauge field; (3) the path integral localises to the fixed points of a supersymmetry, and the localisation formula is an index theorem. These three applications connect the deepest mathematical theorem to the most practical calculations in quantum field theory.

P1. (Zero modes of the Dirac operator.)

The Dirac operator D on a Euclidean 4-manifold M with a gauge field (connection A on a bundle $E \to M$) is: $D_A = \sum_\mu \gamma^\mu(\partial_\mu + A_\mu)$.

(a) A **zero mode** is a solution ψ with $D_A\psi = 0$. By the index theorem:

$$\mathrm{ind}(D_A) = n_+ - n_- = \hat{A}(M) \cdot \mathrm{ch}(E)[M]$$

where $n_\pm$ are the numbers of positive/negative chirality zero modes.

(b) For $M = \mathbb{R}^4$ (non-compact, but with BPST instanton background of instanton number k): the index theorem gives $n_+ - n_- = k$ for a Dirac fermion in the fundamental representation of $SU(2)$. For $k = 1$: there is exactly 1 zero mode (positive chirality), 0 negative chirality zero modes.

(c) The zero modes of D_A contribute to the path integral by the **index density**: the fermion path integral gives a factor of $\det D_A$ which vanishes if $\ker D_A \neq 0$. The instanton background forces certain correlation functions to vanish unless they absorb the zero modes.

(d) **Physical consequence**: in QCD, the instanton zero modes (via the Atiyah-Singer theorem) are responsible for the breaking of the axial $U(1)$ symmetry the resolution of the $U(1)_A$ problem (why the η' meson is heavy). This is a topological effect.

P2. (The chiral anomaly as an index.)

The **chiral anomaly** is the failure of the axial current $J_5^\mu = \bar{\psi}\gamma^\mu\gamma_5\psi$ to be conserved in the presence of gauge fields:

$$\partial_\mu J_5^\mu = \frac{1}{16\pi^2}\mathrm{Tr}(F \wedge F).$$

(a) Integrate over spacetime: $\int_M \partial_\mu J_5^\mu \, d^4x = \frac{1}{16\pi^2}\int_M \mathrm{Tr}(F \wedge F) = 2k$ where k is the instanton number.

(b) The left-hand side is $n_+ - n_-$ by the path integral (the number of positive minus negative chirality zero modes). Conclude: $n_+ - n_- = 2k$ the Atiyah-Singer index theorem!

(c) The chiral anomaly is the index theorem in disguise: the failure of a classical symmetry (chiral symmetry) to survive quantization is measured by the topological charge of the gauge field.

(d) This is the connection between Set 14 (characteristic classes and anomalies) and Set 17 (index theorem): anomaly = index = integral of characteristic class.

P3. (Supersymmetric localisation.)

In a supersymmetric theory, the path integral localises to the fixed points of the supersymmetry. For a supersymmetric quantum mechanics on a manifold M:

(a) The Witten index $Z = \mathrm{Tr}[(-1)^F e^{-\beta H}]$ localises as $\beta \to \infty$ to the zero modes of $H = D^2$, i.e. to $\ker D$.

(b) But Z is also independent of β, so it equals its value at $\beta = 0$ (all modes contribute, but superpartner pairs cancel).

(c) At $\beta = 0$ the path integral reduces to a finite-dimensional integral over the "classical" contributions which is exactly the integral of characteristic classes (the topological index).

(d) This is the physics proof of the Atiyah-Singer theorem: the path integral independence of β forces $\mathrm{ind}(D) = \int_M \hat{A}(TM) \wedge \mathrm{ch}(E)$. The index theorem is a consequence of supersymmetric localisation.

P4. (The complete unification.)

The Atiyah-Singer theorem unifies everything we have done:

Special case	**Operator D**	**Index**	**Sets**
Gauss-Bonnet	$d + d^*$	$\chi(M)$	6B, 12.5
Hirzebruch	signature op.	$\sigma(M)$	14
Riemann-Roch	$\bar{\partial} + \bar{\partial}^*$	$\chi(\mathcal{O})$	15
Witten index	$d + d^*$ (SQFT)	$\chi(M)$	10
Dirac index	Dirac op.	$\hat{A}(M)$	11, 12
Chiral anomaly	Dirac op.	instanton #	14, 16

(a) In every row, the same structure appears: analytical data (kernel/cokernel of a differential operator) equals topological data (integral of characteristic classes). Write a one-sentence statement of why this is profound.

(b) The proof of the index theorem uses heat kernels: $\mathrm{ind}(D) = \mathrm{Tr}(e^{-tD^*D}) - \mathrm{Tr}(e^{-tDD^*})$ is independent of t, and at small t the right-hand side reduces to the integral of characteristic classes via the heat kernel expansion from Set 12.5 P1. Write one sentence connecting the heat kernel (Set 12.5) to the Witten index (Set 10) to the index theorem (Set 17).

(c) **The grand conclusion of the entire course.** We started with a set X and a function $f : X \to X$. We added topology, smooth structure, differential forms, Lie groups, fiber bundles, connections, Riemannian metrics, and symplectic forms. At each step, a new structure answered the questions the previous one could not. The Atiyah-Singer index theorem is the endpoint of this journey: it shows that the deepest question about a differential operator (the dimension of its solution space) is answered by topology (the integrals of characteristic classes). Analysis and topology, local and global, differential equations and algebraic invariants they are the same. Write one paragraph on what you think the next layer of mathematical structure beyond the index theorem might be, and what physical or mathematical question it would answer.

On the left: the analytical index a number computed by solving a differential equation and counting solutions. On the right: a topological index an integral of characteristic classes, computed from the curvature of a connection.

This equality is not obvious. It says that the *number of solutions* to a differential equation is a topological invariant it cannot change under continuous deformations of the equation. Analysis is topology.

The special cases unify everything: Gauss-Bonnet, Riemann-Roch, the signature theorem, the Witten index, the chiral anomaly. All of them are the same theorem.

And the proof via heat kernels and supersymmetric localisation shows that the theorem is a consequence of the deepest symmetry: the pairing between bosons and fermions, between Ω^{even} and Ω^{odd}, between S^+ and S^-.

This is where the course ends. Not because there is nothing more to say but because from here, you can begin to read the literature yourself. Every major development in mathematical physics in the last forty years mirror symmetry, geometric Langlands, M-theory, quantum gravity uses the ideas we have built.

You now have the foundation.***Final Reflection.*** *The Atiyah-Singer index theorem:*

$$ind(D) = \int_M \hat{A}(TM) \wedge ch(E).$$

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